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# Part C

Understanding the forces and moments at the joints while walking is necessary for creating a prosthetic. The prosthetic should hold the maximum force to ensure the user's walking safety. All three joints should be calculated together because one force affects the other, meaning that the force and moments change with each step. Therefore, careful analysis of the human body while walking is necessary to create a prosthetic to assist with walking. Hence, the project aims to learn more about the biomechanical movements of the human body for future innovations.

The data was collected by motion capture, force plate, and anthropometry. The motion capture records the movement data, meaning it obtains the position and orientation of the specific human body points. The force plate captures the ground reaction force exerted by the person. This device measures the accelerations, reactions, and imbalances. Using anthropometry, the device measures the body size and shape, meaning the height, weight, body circumferences, etc. MATLAB was used to calculate the forces at each point.

The x and z values of the forces in the data collected were deciphered to get their components. The ground force reaction (GRF) x and z components were used to get information about the reaction forces. The force plate collected the weight, making calculating the mass easy. The mass would then be used to calculate the right-hand side of the equations of motion. It also made it easier to get the acceleration components of each ligament of the body: foot, shank, and thigh. Not only that, but the center of pressure is also calculated using the force plate. The combination of the devices also leads to calculating the angular accelerations of each ligament and joint. Therefore, the necessary step to solve the equations was to plug in the values of each variable to get the x and z reaction forces for each joint and ligament. After, the moment could be calculated using the distances of the x and z components of the joints and ligaments. After solving one moment, it would be used again to solve the moment for the next joint to see the continuation of the effect of the moment of the initial moment.

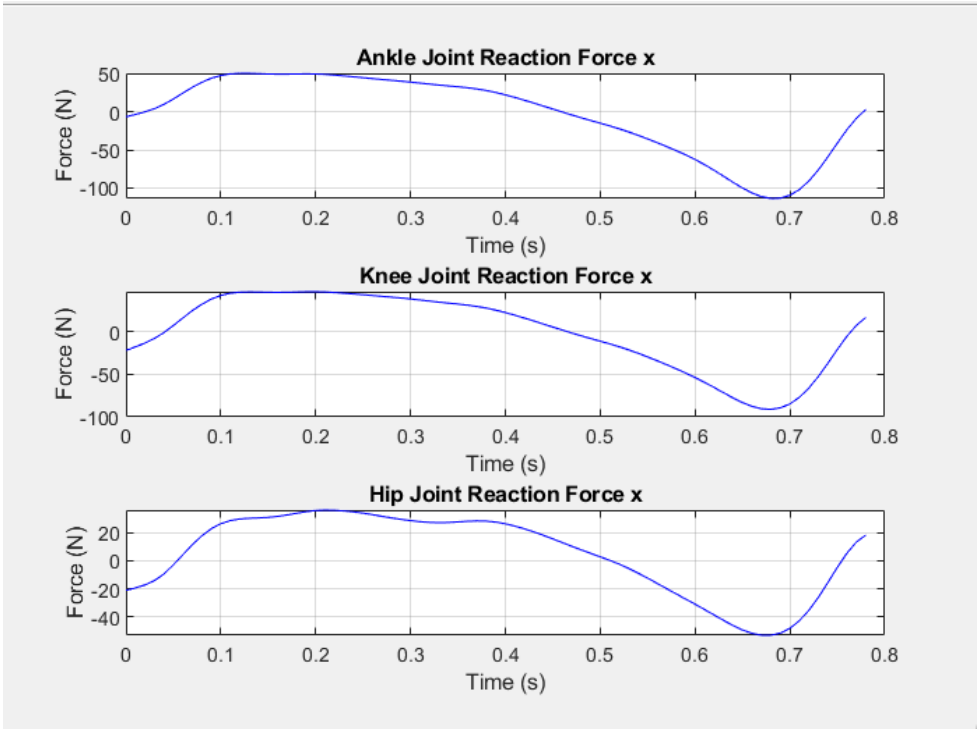



*Figure 1: The x component of the reaction forces of each joint*

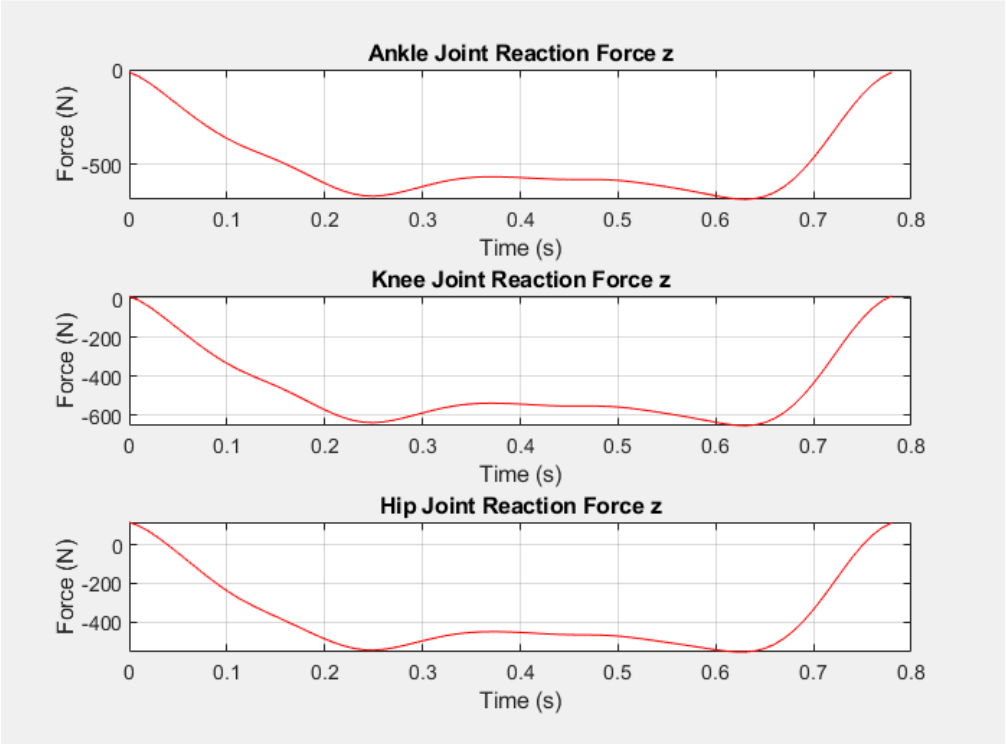



*Figure 2: The z component of the reaction forces of each joint*

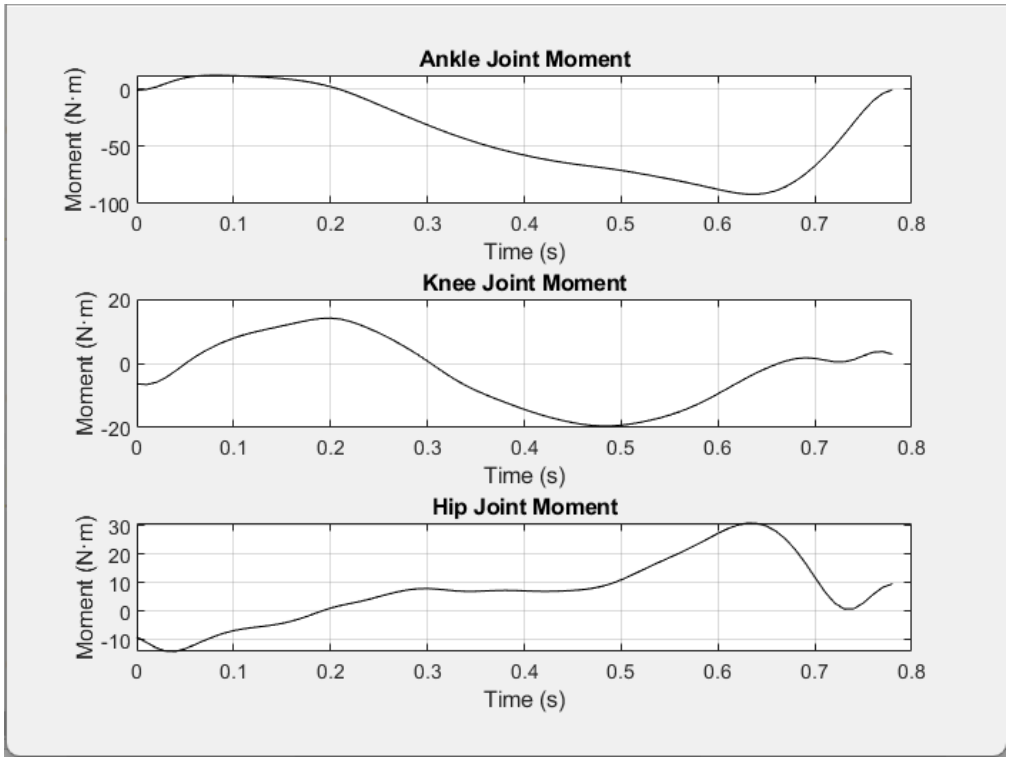



*Figure 3: The moment of each joint.*

Figure 1 shows the x component of the reaction forces from each joint, with the peak force of the ankle receiving the largest force and the peak force of the hip receiving the least when walking forward. Figure 2 shows the z component of the reaction forces from each joint, with the knee receiving the largest peak force and the hip receiving the least peak force when walking forward. Figure 3 shows the moment of each joint, with the ankle receiving the most peak force for the rotation and the knee receiving the least when walking forward.

The ankle has the most significant magnitude for the reaction force in the z direction due to the amount of weight it has to carry. It must support the shank, thigh, and body, which is more weight than the knee or hip that needs support. This peak occurs at the terminal stance, when the person's heel is off the ground, and the center of pressure is on the front of the foot. During the terminal stance, only one foot/leg supports the person. This happens so they can swing the other leg forward. Hence, the reason for the peaks in the moment of each joint. The reaction forces of the joints on the other leg are much lower since they no longer support the person's weight.

The x direction has the same reason as the z direction since the ankle carries a higher load. The force is positive due to the person's movement; the force in the x direction should be positive since they go forward. However, as it went to the terminal stance, it went negative due to the push of that leg to swing it after the terminal stance. It also kept the person from falling as they moved forward.

The moment was another figure that was calculated. In the reference, when calculated, counterclockwise was positive, meaning that the opposite direction is negative. It could also be

considered the extension for the counterclockwise motion and the bending of the clockwise motion. When the person walked forward, the ankle joint started counterclockwise; however, as time passed, the ankle moment went clockwise. On the contrary, the hip moment is positive. Therefore, the hip moved counterclockwise or extending, which, according to the reference frame, would lead to a positive number when calculated. When the ankle and hip moments are compared, the difference is that the ankle moment is negative; however, the hip moment is positive. There is also more force on the ankle moment, even though it is in the negative direction. The cause for this discrepancy is the ankle joint went clockwise while the hip went counterclockwise. Also, the force difference came from the amount of weight the ankle can hold. Yet, the hip does not carry the person's body weight, meaning that the amount of force the hip needs to move is lower. Not only that, but the amount of force absorbed by the limb lowers the amount of force that the hip has to move.

According to Figure 3, the moment of the knee had two extremas. The initial went positive to negative, meaning this was the initial contact to the mid-stance. This was because the knee joint at the initial contact was negative, then rotated counterclockwise, meaning it started to extend. The leg started to squeeze after the mid-stance, meaning it moved clockwise and was negative. Near the end, it was shown that it went back to 0 Nm due to the extension of the leg. Overall, the slope went positive, negative, and positive to showcase the extension and bending of the leg.